

Contribution Journal

Wine Classification — Machine Learning Workflow

COURSE	ITAI 1371 — Classic Machine Learning
ASSIGNMENT	Module 03 Lab Exercise, Contribution Journal
SUBJECT	Wine Classification — Supervised Learning Models
GROUP	SingleEpoch (solo member — this journal covers 100% of the work)
AUTHOR	Joseph Clay

1. Team Structure

I completed this assignment independently as the sole member of my group, SingleEpoch. Every section below describes my own work from start to finish, with no other contributors to credit or divide work between.

2. What I Built

I worked through the full machine learning workflow on the Wine dataset: loading and exploring the data, visualizing class distribution and feature correlations, selecting features, splitting the data into training and test sets, scaling the features correctly, training a Logistic Regression model and a Decision Tree, evaluating both, and interpreting the results with a confusion matrix.

3. How I Worked

Rather than writing the whole notebook and running it once at the end, I moved through it one step at a time and checked each result before continuing. For nearly every step, I added a real check into the code itself instead of just reading the printed output and assuming it was right. For example, I confirmed the dataset actually had 178 samples across 3 classes before doing anything else with it, confirmed the training and test split kept every class properly represented, and confirmed the scaler was fit only on the training data before it ever touched the test set. Where two numbers should have matched, such as the accuracy printed after evaluation and the accuracy the confusion matrix implies, I checked that they actually did rather than assuming they would.

4. The Learning Curve — My Main Addition Beyond the Assignment

The lab's original evaluation step only reports one test-set accuracy number per model. I added a learning curve analysis on top of that: for each model, I trained it repeatedly on progressively larger portions of the training data and compared its training accuracy against its cross-validation accuracy at each size. Plotting both lines together shows directly whether a model is overfitting, meaning its training accuracy sits far above its validation accuracy, or genuinely learning, meaning the two lines converge as more data is added.

Both of my models showed small, healthy gaps: 0.035 for Logistic Regression and 0.038 for the Decision Tree. That is the strongest piece of evidence in this project that the 91.7% accuracy result is real and not a lucky split. It also let me correct an assumption I made early in my own reflection. I had first guessed the Decision Tree scored lower because it overfit, but the learning curve showed that was not true, so I went back and rewrote that explanation to match the actual evidence instead of my initial guess.

5. Keeping Notes as I Went

At each stage I wrote down, in my own words, what the step was for, what it produced, and the underlying concept behind it, instead of trying to reconstruct that understanding after the fact. That habit is what made the final reflection straightforward to write. Most of the key insights were already drafted from actually building each piece, not written cold at the end.